



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 8.7.24

Instructions:

Attempt all questions. The intended marks for questions are given in brackets []. This paper consists of three printed pages.

Question 1

Choose the correct answer from the choices given:

[6×1=6]

- i) The process of encapsulating a value within an object and changing a primitive type into a wrapper class is known as _____
- a) converting b) unboxing c) wrapping d) boxing
- ii) What is the output of the given expression:
- ```
int a= -1, b=0, c=3;
System.out.print(--a - b++ * c-- %2);
```
- a) -3      b) -2      c) 0      d) -1
- iii) Assertion(A): If the number of iteration is fixed, it is recommended to use the for loop.  
Reason(R): The number of iterations are not fixed, it is advisable to use while or do while loop.
- (a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)  
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)  
(c) Assertion (A) is true and Reason (R) is false  
(d) Assertion (A) is false and Reason(R) is true
- iv) \_\_\_\_\_ method uses compile time or early binding whereas \_\_\_\_\_ method uses runtime or dynamic binding.
- a) static, non static      b) pure, impure      c) non static, static      d) impure, pure
- v) Wrapper class are present in \_\_\_\_\_ package.
- a) java.io      b) java.lang      c) java.util      d) java.wrapper

vi) What is the final value of 'ctr' after the iteration process given below executes?

```
int ctr=0;
for(int i=1;i<=5;i++)
ctr++;
for(int j=1;j<=5;j+=2)
++ctr;
```

- a) 6                      b) 8                      c) 7                      d) 9

#### Question 2

a) Write a statement in java to convert a number stored in a string variable 'n' to double data type. [2]

b) Predict the output: [2]

```
class pg1
{ static public int fun(int x, int y)
{ y=x; return y; }
public static void main()
{ int a=3, b=7;
b=fun(a,b);
a=fun(b,a);
System.out.println(a + " " +b); } }
```

c) Predict the output: [2]

```
class abc
{ void main()
{ Integer n=new Integer(235);
String s= "235";
int p=Integer.parseInt(s);
p=p+5;
System.out.println(f);
System.out.println(p); }}
```

d) Show the difference between call by value and call by reference with the help of an example? [2]

e) What will be the final value of 'p', when x = -2, y = 4: [2]

```
p = Math.pow(x++ + ++x - --y + x--, --y)
```

f) Convert the given code snippet to equivalent if else construct:

[2]

```
switch(obj)

{ case 'a': System.out.print("Platform Independent");

 case 'b': System.out.print("object oriented");

 default: System.out.print("wrong choice"); }
```

g) The following program segment calculates the norm of a number, norm of a number is the square root of sum of squares of all digits of a number.

For example: The norm of 68 is 10.  $(6 \times 6 + 8 \times 8) = 100$ , square root of 100 is 10.

Fill in the blanks with appropriate java statements:

[2]

```
void norm(int n)

{ int d, s=___ ;

 while(____)

 { d=n%10;

 s = ____;

 n=n/10; }

 System.out.print("norm =" + ____); }
```

### Question 3

[10]

Design a class to overload the method display(.....) as follows:

1. void display( ) — prints the 10<sup>th</sup> character after the alphabetical letter 'n'.
2. void display(char s, char c) — generate all the alternate letters from 's' to 'c'.
3. int display(char st) — checks and print whether 'st' is a special characters or not. Return 1 if 'st' is a special character otherwise return 0.

#### Question 4

[10]

Design a menu-driven program to display the following as per the user's choice. Accepting the choice as 'A' to check for Abundant Number and 'B' to check for ugly number.

- (a) An Abundant number is a number for which the sum of its proper factors is greater than the number itself.

Example: Consider the number 12. Factors of 12 = 1, 2, 3, 4, 6

Sum of factors =  $1 + 2 + 3 + 4 + 6 = 16$ .

As  $16 > 12$  so 12 is an Abundant number.

- (b) An ugly number is a positive integer whose prime factors are limited to 2, 3, and 5.

Example 1: Input n = 6

Output = true

Explanation:  $6=2*3$

Example 2: Input n = 14

Output = false

Explanation: 14 includes 7 as prime factor